

Exercise J02

Q1. A is a square matrix of order n.

l = maximum number of distinct entries if A is a triangular matrix

m = maximum number of distinct entries if A is a diagonal matrix

p = minimum number of zeroes if A is a triangular matrix

If $l + 5 = p + 2m$, find the order of the matrix.

Ans. 4

Sol. $l = 1 + 2 + 3 + \dots + n \Rightarrow l = \frac{n(n+1)}{2} + 1$;

$$m = n + 1 \quad p = \frac{n(n-1)}{2}$$

Q2. If, $E = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and $F = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ calculate the matrix product EF & FE and show that $E^2F + FE^2 = E$.

Ans. $EF = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $FE = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Sol. -

Q3. Find the value of x and y that satisfy the equations.

$$\begin{bmatrix} 3 & -2 \\ 3 & 0 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x & y \\ x & x \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3y & 3y \\ 10 & 10 \end{bmatrix}$$

Ans. $x = \frac{3}{2}$, $y = 2$

Sol. -

Q4. If, $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$, then show that the matrix A is a root of the polynomial $f(x) = x^3 - 6x^2 + 7x + 2$.

Ans. -

Sol. -

Q5. Consider the two matrices A and B where $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$; $B = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$. If n(A) denotes the number of elements in A such that $n(XY) = 0$, when the two matrices X and Y are not

conformable for multiplication. If $C = (AB)(B'A)$; $D = (B'A)(AB)$ then, find the value of $\left(\frac{n(C)(|D|^2 + n(D))}{n(A) - n(B)} \right)$

Ans. 650

Sol. $n(A) = 4$; $n(B) = 2$; $n(C) = 4$; $n(D) = 1$; $|D| = 18$

$$\frac{n(C)(|D|^2 + n(D))}{n(A) - n(B)} = \frac{4(18^2 + 1)}{2} = 650$$

Q6. define $A = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$. Find a vertical vector V such that $(A^8 + A^6 + A^4 + A^2 + I)V = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$ (where I is the 2×2 identity matrix).

Ans. $v = v = \begin{bmatrix} 0 \\ \frac{1}{11} \end{bmatrix}$

Sol. $A = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$; $A^2 = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = 3I$

$$(1 + 3 + 3^2 + 3^3 + 3^4) I \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$$

$$(121) \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 11 \end{bmatrix} \Rightarrow \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 1/11 \end{bmatrix}$$

Q7. If $\begin{bmatrix} a & b \\ c & 1-a \end{bmatrix}$ is an idempotent matrix. Find the value of $f(a)$, where $f(x) = x - x^2$, when $bc = 1/4$. Hence otherwise evaluate a .

Ans. $f(a) = 1/4$, $a = 1/2$

Sol. -

Q8. If the matrix A is involutory, show that $\frac{1}{2}(I + A)$ and $\frac{1}{2}(I - A)$ are idempotent and $\frac{1}{2}(I + A) \cdot \frac{1}{2}(I - A) = O$.

Ans. -

Sol. -

Q9. Show that the matrix $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ can be decomposed as a sum of a unit and a

nilpotent matrix. Hence evaluate the matrix $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}^{2007}$.

Ans. $\begin{bmatrix} 1 & 0 \\ 4014 & 1 \end{bmatrix}$

Sol. -

Q10. Let X be the solution set of the equation $Ax = I$, where $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ and I is the

corresponding unit matrix and $x \subseteq \mathbb{N}$ then find the minimum value of

$$\sum (\cos^x \theta + \sin \theta), \theta \in R.$$

Ans. 2

Sol. $A^2 = I \Rightarrow x = 2, 4, 6, \dots$

$$\sum (\cos^x \theta + \sin^x \theta)$$

$$= (\cos^2 \theta + \sin^2 \theta) + (\cos^4 \theta + \sin^4 \theta) + \dots$$

$$= (\cot^2 \theta + \tan^2 \theta) \leq 2$$

Q11. If A is an idempotent non zero matrix and I is an identity matrix of the same order, find the value of n, $n \in \mathbb{N}$, such that $(A + I)^n = I + 127 A$.

Ans. $n = 7$

Sol. -

Q12. Given matrices $A = \begin{bmatrix} 1 & x & 1 \\ x & 2 & y \\ 1 & y & 3 \end{bmatrix}$; $B = \begin{bmatrix} 3 & -3 & z \\ -3 & 2 & -3 \\ z & -3 & 1 \end{bmatrix}$ Obtain x, y and z if the matrix AB

is symmetric.

Ans. $\left(-\frac{4\sqrt{2}}{3}, \frac{2}{3}, 2\sqrt{2}\right), \left(\frac{4\sqrt{2}}{3}, \frac{2}{3}, -2\sqrt{2}\right), (3, 3, -1)$

Sol. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix} \Rightarrow |A| = 6$

$$A^{-1} \Rightarrow \frac{adjA}{|A|} = \frac{1}{6} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 2 & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 6 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{bmatrix}$$

$$A^{-1} = \frac{1}{6} [A^2 + cA + dI]$$

$$\frac{1}{6} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & -2 & 1 \end{bmatrix}$$

$$= \frac{1}{6} \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 5 \\ 0 & -10 & 14 \end{bmatrix} + \begin{bmatrix} c & 0 & 0 \\ 0 & c & c \\ 0 & -2c & 4c \end{bmatrix} \begin{bmatrix} d & 0 & 0 \\ 0 & d & 0 \\ 0 & 0 & d \end{bmatrix} \right\}$$

on comparing we get

$$1 = 5 + c \Rightarrow c = -6$$

$$1 = 14 + 4c + d \Rightarrow 1 = 14 - 24 + d$$

$$d = 11$$

Q13. $A = \begin{pmatrix} 3 & a & -1 \\ 2 & 5 & c \\ b & 8 & 2 \end{pmatrix}$ is Symmetric and $B = \begin{pmatrix} d & 3 & a \\ b-a & e & -2b-c \\ -2 & 6 & -f \end{pmatrix}$ is Skew Symmetric, then find AB.

Is AB a symmetric, Skew Symmetric or neither of them. Justify your answer.

Ans. AB is neither symmetric nor skew symmetric

Sol. -

Q14. Express the matrix $\begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & -6 \\ -1 & 0 & 4 \end{bmatrix}$ as a sum of a lower triangular matrix & an upper

triangular matrix with zero in its leading diagonal. Also Express the matrix as a sum of a symmetric & a skew symmetric matrix

$$\text{Ans. } \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ -1 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 5 \\ 0 & 0 & -6 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 1 & 2 & 2 \\ 2 & 3 & -3 \\ 2 & -3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & -3 \\ -3 & 3 & 0 \end{bmatrix}$$

Sol. -

Q15. Let $A = [a_{ij}]_{3 \times 3}$ be a matrix. If $A + A^T = \begin{bmatrix} 6 & 4 & 4 \\ a_{21} + a_{12} & 10 & a_{23} + a_{32} \\ a_{31} + a_{13} & 4 & 8 \end{bmatrix}$ where a_{12}, a_{23} and a_{31} are the positive root of the equation $x^3 - 6x^2 + px - 8 = 0$, $p \in \mathbb{R}$, then the value of $\det(A)$ is [Note : A^T denotes the transpose of matrix A .]

Ans. 28

Sol. Clearly $A + A^T$ is a symmetric matrix

Also, equation $x^3 - 6x^2 + 8x - 8 = 0$ has roots a_{12}, a_{23}, a_{31}

$$\therefore a_{12} + a_{23} + a_{31} = 8$$

$$A.M. = G.M. \Rightarrow a_{12} = a_{23} = a_{31} = 2$$

$$\therefore \left. \begin{array}{l} a_{21} + a_{12} = 4 \\ a_{31} + a_{13} = 4 \\ a_{23} + a_{32} = 4 \end{array} \right\} \Rightarrow a_{21} = a_{13} = a_{23} = 2$$

and for diagonal elements.

$$2a_{11} = 6, 2a_{22} = 10, 2a_{33} = 8$$

$$\Rightarrow a_{11} = 3, a_{22} = 5, a_{33} = 4$$

$$\therefore A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 5 & 2 \\ 2 & 2 & 4 \end{bmatrix}$$

$$\Rightarrow |A| = 3(20 - 4) - 2(8 - 1) + 2(4 - 10)$$

$$= 48 - 8 - 12 = 28$$

Q16. If M is a 3×3 matrix, where $M^T M = I$ and $\det(M) = 1$, then prove that $\det(M - I) = 0$.

Ans. -

$$\text{Sol. } |M - I| = |M - M M^T|$$

$$|M - I| = |M| |I - M|$$

$$\Rightarrow |M - I| = |I - M|$$

$$\Rightarrow |M - I| = (-1)^3 |M - I|$$

$$\Rightarrow |M - I| = 0$$

Q17. If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ then the trace of the matrix $\text{Adj}(\text{Adj } A)$ is

Ans. 1

Sol. -

Q18. If $A = \begin{bmatrix} u & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$, where $\left(u \neq -\frac{11}{3}\right)$ and $\det(\text{adj } A) = (23)^4$, then the value of u is

Ans. 4

Sol. -

Q19. Let A be a 3×3 matrix such that $a_{11} = a_{33} = 2$ and all the other $a_{ij} = 1$. Let $A^{-1} = xA^2 + yA + zI$ then find the value of $(x + y + z)$ where I is a unit matrix of order 3.

Ans. 1

Sol.

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$A^{-1} = xA^2 + yA + zI$$

$$\Rightarrow xA^3 + yA^2 + zA - I = 0$$

is characteristic equation of A

$$\begin{vmatrix} 2 - \lambda & 1 & 1 \\ 1 & 1 - \lambda & 1 \\ 1 & 1 & 2 - \lambda \end{vmatrix} = 0$$

$$\lambda^2 - 5\lambda^2 + 5\lambda - 1 = 0$$

$$\Rightarrow x + y + z = 1$$

Q20. For the matrix $A = \begin{bmatrix} 4 & -4 & 5 \\ -2 & 3 & -3 \\ 3 & -3 & 4 \end{bmatrix}$ find A^{-2} .

Ans. $\begin{bmatrix} 17 & 4 & -19 \\ -10 & 0 & 13 \\ -21 & -3 & 25 \end{bmatrix}$

Sol. -

Q21. If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then show that $F(x) \cdot F(y) = F(x + y)$

Hence prove that $[F(x)]^{-1} = F(-x)$.

Ans. -

Sol. -

Q22. Let M be a 3×3 matrix satisfying $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$, $M \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ and $M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$.

Then the sum of the diagonal entries of M is

Ans. 9

Sol. Let $M = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$$\text{then } a_{12} = -1, \quad a_{11} - a_{12} = 1 \quad \Rightarrow \quad a_{11} = 0,$$

$$a_{22} = 2, \quad a_{21} - a_{22} = 1 \quad \Rightarrow \quad a_{21} = 3,$$

$$a_{32} = 3, \quad a_{31} - a_{32} = -1 \quad \Rightarrow \quad a_{31} = 2,$$

Hence sum of diagonal of M is $= a_{11} + a_{22} + a_{33} = 0 + 2 + 7 = 9$

Q23. Given $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$. Find P such that $BPA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

Ans. $\begin{bmatrix} -4 & 7 & -7 \\ 3 & -5 & 5 \end{bmatrix}$

Sol. -

Q24. Use matrix to solve the following system of equations.

$$\begin{array}{llll} \text{(i) } x + y + z = 3 & \text{(ii) } x + y + z = 6 & \text{(iii) } x + y + z = 3 & \text{(iv) } x + y + z = 3 \\ x + 2y + 3z = 4 & x - y + z = 2 & x + 2y + 3z = 4 & x + 2y + 3z = 4 \\ x + 4y + 9z = 6 & 2x + y - z = 1 & 2x + 3y + 4z = 7 & 2x + 3y + 4z = 9 \end{array}$$

Ans. (i) $x = 2, y = 1, z = 0$; (ii) $x = 1, y = 2, z = 3$;
(iii) $x = 2 + k, y = 1 - 2k, z = k$ where $k \in \mathbb{R}$; (iv) inconsistent, hence no solution

Sol. -

$$\begin{array}{l} \text{Q25. } x + y + z = 6 \\ x - y + z = 2 \\ 2x + y - z = 1 \end{array}$$

Ans. $x = 1, y = 2, z = 3$

Sol. We have,

$$x + y + z = 6, x - y + z = 2, 2x + y - z = 1$$

$$\text{Thsu } A = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix}$$

$$X = \begin{vmatrix} x \\ y \\ z \end{vmatrix}, B = \begin{vmatrix} 6 \\ 2 \\ 1 \end{vmatrix}$$

Now solve A to get A = 6

$$\text{Now } x = \frac{A_x}{A} = \frac{\begin{vmatrix} 6 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix}}$$

$$y = \frac{A_y}{A} = \frac{\begin{vmatrix} 1 & 6 & 1 \\ 1 & 2 & 1 \\ 2 & 1 & -1 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix}} = \frac{12}{6} = 2$$

$$z = \frac{A_z}{A} = \frac{\begin{vmatrix} 1 & 1 & 6 \\ 1 & -1 & 2 \\ 2 & 1 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix}} = \frac{18}{6} = 3$$

Thus $x = 1, y = 2, z = 3$

Q26. $x + y + z = 3$

$$x + 2y + 3z = 4$$

$$2x + 3y + 4z = 7$$

Ans. $x = 2+k, y = 1-2k, z = k$ where $k \in \mathbb{R}$

Sol. Given system of equations can be written as $AX = B$

where $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; \quad B = \begin{bmatrix} 3 \\ 4 \\ 7 \end{bmatrix}$$

Here, $|A| = 0$

Now, we will find $(\text{adj}A)B$

$$\text{adj } A = C^T = \begin{bmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ 1 & -2 & 1 \end{bmatrix}^T$$

$$\Rightarrow \text{adj}A = \begin{bmatrix} -1 & -1 & 1 \\ 2 & 2 & -2 \\ -1 & -1 & 1 \end{bmatrix}$$

$$\text{Now, } (\text{adj}A)B = \begin{bmatrix} -1 & -1 & 1 \\ 2 & 2 & -2 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 7 \end{bmatrix}$$

$$\Rightarrow (\text{adj}A)B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow (\text{adj}A)B = 0$$

Hence, the system of equations has infinitely many solutions.

Let $z = k$ where $k \in \mathbb{R}$

Then

$$x + y = 3 - k$$

$$x + 2y = 4 - 3k$$

Solving these eqns, we get

$$y = 1 - 2k; \quad x = 2 + k$$

Q27. $x + y + z = 3$

$$x + 2y + 3z = 4$$

$$2x + 3y + 4z = 9$$

Ans. inconsistent, hence no solution

Sol.

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{vmatrix} = 1(8 - 9) - 1(4 - 6) + 1(3 - 4) \\ = -1 + 2 - 1 = 0$$

Paragraph

Let A and B are two matrices of same order i.e. where $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & K & 5 \\ 4 & 2 & 1 \end{bmatrix}$ B =

$$\begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 4 \\ 3 & 3 & 5 \end{bmatrix}$$

Q28. If A is singular matrix then $\text{tr}(A + B)$ is equal to :

(A) 5 (B) 3 (C) 4 (D) 6

Ans. B

Sol.

$$|A| = 0$$

$$k - 10 + 3(2 - 20) + 2(4 - 4k) = 0$$

$$k - 10 - 54 + 8 - 8k = 0$$

$$7k = -56$$

$$k = -8$$

$$\text{tr}(A + B) = -6 + 9 = 3$$

Paragraph

Let A and B are two matrices of same order i.e. where $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & K & 5 \\ 4 & 2 & 1 \end{bmatrix}$ B =

$$\begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 4 \\ 3 & 3 & 5 \end{bmatrix}$$

Q29. If $K = 2$ then $\text{tr}(AB) + \text{tr}(BA)$ is equal to :

(A) 66 (B) 42 (C) 84 (D) 63

Ans. C

Sol.

$$AB = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 2 & 5 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 4 \\ 3 & 3 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & - & - \\ - & 21 & - \\ - & - & 25 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 4 \\ 3 & 3 & 5 \end{bmatrix} \begin{bmatrix} 1 & -3 & 2 \\ 2 & 2 & 5 \\ 4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 16 & - & - \\ - & 0 & - \\ - & - & 26 \end{bmatrix}$$

$$\text{tr}(AB) + \text{tr}(BA) = 42 + 42 = 84$$

Paragraph

Let A and B are two matrices of same order i.e. where $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & K & 5 \\ 4 & 2 & 1 \end{bmatrix}$ B =

$$\begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 4 \\ 3 & 3 & 5 \end{bmatrix}$$

Q30. If $C = A - B$ and $\text{tr}(C) = 0$ then K is equal to :

(A) 5 (B) -5 (C) 7 (D) -7

Ans. C

$$\text{Sol. } C = \begin{bmatrix} -1 & -4 & -1 \\ -2 & k-2 & 1 \\ 1 & -1 & -4 \end{bmatrix}$$

$$\text{tr}(C) = 0$$

$$-7 + k = 0$$

$$k = 7$$

Paragraph

Find the rank of the following matrices

\n

$$\text{Q31. } \begin{bmatrix} 2 & 3 & 5 \\ 1 & 1 & 6 \\ 1 & 4 & 3 \end{bmatrix}$$

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. D

Sol. -

Paragraph

Find the rank of the following matrices

\n

Q32.
$$\begin{bmatrix} 1 & 3 & -2 \\ 4 & -1 & 0 \\ 2 & -7 & 4 \end{bmatrix}$$

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. -

Paragraph

Find the rank of the following matrices

\n

Q33.
$$\begin{bmatrix} 2 & -1 & 3 & 5 \\ 1 & 4 & 0 & 2 \\ 0 & -9 & 3 & 1 \\ 3 & 3 & 3 & 7 \end{bmatrix}$$

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. -

Paragraph

$A = \begin{bmatrix} \text{\textcolor{red}{n1}} & 0 & 0 \\ \text{\textcolor{red}{n2}} & 1 & 0 \\ \text{\textcolor{red}{n3}} & 2 & 1 \end{bmatrix}$, U_1, U_2 and U_3 are columns matrices satisfying. $AU_1 = \begin{bmatrix} \text{\textcolor{red}{n1}} \\ \text{\textcolor{red}{n0}} \\ \text{\textcolor{red}{n0}} \end{bmatrix}$; $AU_2 = \begin{bmatrix} \text{\textcolor{red}{n2}} \\ \text{\textcolor{red}{n3}} \\ \text{\textcolor{red}{n0}} \end{bmatrix}$, $AU_3 = \begin{bmatrix} \text{\textcolor{red}{n2}} \\ \text{\textcolor{red}{n3}} \\ \text{\textcolor{red}{n1}} \end{bmatrix}$ and U is 3×3 matrix whose columns are U_1, U_2, U_3 then answer the following questions

\n

Q34. The value of $|U|$ is

- (A) 3 (B) -3 (C) $3/2$ (D) 2

Ans. A

Sol. Let $U_1 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ then $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a \\ 2a + b \\ 3a + 2b + c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow a = 1, \text{ then } b = -2, c = 1$

$\therefore U_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ Similarly, $U_2 = \begin{bmatrix} 2 \\ -1 \\ -4 \end{bmatrix}, U_3 = \begin{bmatrix} 2 \\ -1 \\ -3 \end{bmatrix}$

$\therefore U = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -1 \\ 1 & -4 & -3 \end{bmatrix} \Rightarrow |U| = 3$

Paragraph

$A = \begin{bmatrix} \text{\textcolor{red}{n}1} & 0 & 0 \\ \text{\textcolor{red}{n}2} & 1 & 0 \\ \text{\textcolor{red}{n}3} & 2 & 1 \end{bmatrix}$, U_1, U_2 and U_3 are columns matrices satisfying. $AU_1 = \begin{bmatrix} \text{\textcolor{red}{n}1} \\ \text{\textcolor{red}{n}0} \\ \text{\textcolor{red}{n}0} \end{bmatrix}$; $AU_2 = \begin{bmatrix} \text{\textcolor{red}{n}2} \\ \text{\textcolor{red}{n}3} \\ \text{\textcolor{red}{n}0} \end{bmatrix}$, $AU_3 = \begin{bmatrix} \text{\textcolor{red}{n}2} \\ \text{\textcolor{red}{n}3} \\ \text{\textcolor{red}{n}1} \end{bmatrix}$ and U is 3×3 matrix whose columns are U_1, U_2, U_3 then answer

the following questions

\n

Q35. The sum of elements of U^{-1} is

(A) -1 (B) 0 (C) 1 (D) 3

Ans. B

Sol. $U^{-1} = \frac{1}{3} \begin{bmatrix} -1 & -2 & 0 \\ -7 & -5 & -3 \\ 9 & 6 & 3 \end{bmatrix} \Rightarrow \text{Sum of the elements} = 0$

Paragraph

$A = \begin{bmatrix} \text{\textcolor{red}{n}1} & 0 & 0 \\ \text{\textcolor{red}{n}2} & 1 & 0 \\ \text{\textcolor{red}{n}3} & 2 & 1 \end{bmatrix}$, U_1, U_2 and U_3 are columns matrices satisfying. $AU_1 = \begin{bmatrix} \text{\textcolor{red}{n}1} \\ \text{\textcolor{red}{n}0} \\ \text{\textcolor{red}{n}0} \end{bmatrix}$; $AU_2 = \begin{bmatrix} \text{\textcolor{red}{n}2} \\ \text{\textcolor{red}{n}3} \\ \text{\textcolor{red}{n}0} \end{bmatrix}$, $AU_3 = \begin{bmatrix} \text{\textcolor{red}{n}2} \\ \text{\textcolor{red}{n}3} \\ \text{\textcolor{red}{n}1} \end{bmatrix}$ and U is 3×3 matrix whose columns are U_1, U_2, U_3 then answer

the following questions

\n

Q36. The value of $\begin{bmatrix} 3 & 2 & 0 \end{bmatrix} U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$ is

- (A) 5 (B) $5/2$ (C) 4 (D) $3/2$

Ans. A

Sol.
$$[3 \ 2 \ 0] \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -1 \\ 1 & -4 & -3 \end{bmatrix} = \begin{bmatrix} 1 \\ -8 \\ 0 \end{bmatrix} [3 \ 2 \ 0] \begin{bmatrix} 7 \\ -8 \\ -5 \end{bmatrix} = [5]$$

Q37. If $AB = A$ and $BA = B$, then

- (A) $A^2B = A^2$ (B) $B^2A = B^2$ (C) $ABA = A$ (D) $BAB = B$

Ans. D, C, B, A

Sol. We have $A^2B = A(AB) = AA = A^2$, $B(BA) = BB = B^2$, $ABA = A(BA) = AB = A$, and $BAB = B(AB) = B$

Q38. A square matrix A with elements from the set of real numbers is said to be orthogonal if $A' = A^{-1}$. If A is an orthogonal matrix, then

- (A) A' is orthogonal (B) A^{-1} is orthogonal (C) $\text{Adj. } A = A'$ (D) $|A^{-1}| = 1$

Ans. B

Sol. $A^{-1}A = I \leftrightarrow A'A = I \leftrightarrow (A')'A' = I \Rightarrow A'$ is orthogonal

Also, $(A^{-1})' = (A')^{-1} = (A^{-1})^{-1} = A = A$

$\Rightarrow A^{-1}(A^{-1})' = I$

$\therefore A^{-1}$ is orthogonal

Next $AA' = I \Rightarrow |AA'| = 1 \Rightarrow |A| |A'| = 1$

$\Rightarrow |A|^2 = 1 \Rightarrow |A| = \pm 1$

As $\text{Adj. } A = |A| A^{-1} = |A|A'$, we cannot say, $\text{Adj. } A = A'$

Also if $|A| = 1$, then $|A^{-1}| = \pm 1$

Q39. If A and B are two 3×3 matrices such that their product AB is a null matrix then

(A) $\det. A \neq 0 \implies B$ must be a null matrix.

(B) $\det. B \neq 0 \implies A$ must be a null matrix.

(C) If none of A and B are null matrices then atleast one of the two matrices must be singular.

(D) If neither $\det. A$ nor $\det. B$ is zero then the given statement is not possible.

Ans. D, C, B, A

Sol. $AB = O$

$$\therefore |AB| = 0 \Rightarrow |A| |B| = 0$$

$$\therefore \det A \neq 0$$

$$\therefore A^{-1} \text{ exist}$$

$$IB = 0$$

$$B = 0 \Rightarrow B \text{ must be null matrix}$$